and commercial operators. Much of the early activity of this revolution occurred, and is documented on the DIY Drones website *(https://diydrones.com)*, an online forum for hobbyists interested in developing autonomous model aircraft.

Phase 3: Open source silicon

Although open source technologies gave developers royalty-free access to hardware and software platforms that they could modify to suit their needs, the integrated circuits their platforms run on have been almost exclusively based on proprietary IP. This began to change as devices known as field-programmable gate arrays (FPGAs) became inexpensive enough to be used as the equivalent of a custom integrated circuit. Designers can program the devices with the desired functionality using large libraries of FPGA code (both proprietary and open source) in a matter of days or weeks, as opposed to the months and millions of dollars required to develop the tooling for a real IC.

Although FPGAs are excellent solutions for certain applications, they are slower, more power hungry and significantly more expensive than an equivalent single-purpose silicon device. Until recently, nearly all 'hard' silicon designs were based on proprietary IP, either developed in-house or licensed from a third party vendor. The cost and time required to develop a reliable, high-performance processor architecture meant that nearly all ICs that require a processor core use third-party IP, licensed from a handful of vendors such as ARC, ARM, MIPS and Qualcomm. These cores offer excellent technical support, optional architectural enhancements, and deep libraries of application code, but the IP tends to be very costly to license and usually has significant restrictions on how it can be modified.

As a result, the same market pressures that created open source software and hardware are giving rise to the first generation of open source silicon projects. Among the most ambitious and successful of these efforts is the RISC-V project, an open instruction set architecture (ISA), originally developed at the University of California at Berkeley, and currently supported by the RISC-V Foundation *(https://riscv.org/)*.

Like most modern ISAs, it is based on reduced instruction set computing (RISC) principles that provide a good balance between silicon footprint, execution speed and code efficiency. The instruction set can be instantiated with a 32-, 64-, or 128-bit word width and has been designed to efficiently support a wide variety of common applications. Unlike proprietary ISAs, the RISC-V ISA may be freely used for any purpose, enabling anyone to design, produce or sell RISC-V chips and software.

The Foundation's 100+ member organisations have pooled their knowledge, know-how and resources to create a common platform and an ecosystem of development tools that form the basis of their own unique and highly competitive products. SiFive *(https://www.sifive.com/)*, for example, is developing a service that allows its customers to design custom RISC-V based System on a Chip (SoC) devices and receive sample chips within a matter of weeks. SiFive also enables newcomers to become familiar with the technology by offering a low-cost, Arduino-compatible board powered by one of its RISC-V processors.

RISC-V's customisable architecture makes it easy for developers to create devices with the precise mix of functionality, performance and energy-saving features that give their products the unique capabilities they need to dominate a particular application space. As an example, Trinamic, a German semiconductor maker focused on motor drivers, is adopting the RISC-V as the foundation for its next generation of motor drivers and power electronics. The amount of computational power required for motor control functions has been

Open standards vs open source

Open standards helped make our industrial economy possible. They don't tell a manufacturer anything about how to design or make a product; they define its basic functionality, especially with regard to how it interacts with similar products from other manufacturers. At the dawn of the Industrial Revolution, standards for things like the thread pitches of bolts and screws, the calibre of bullets, and railway cars' wheel spacing made them easier and less expensive to manufacture, thereby creating a broader demand for them.

In more recent times, open standards have allowed products ranging from mobile phones and wireless LAN components, to batteries and computer memories enjoy the benefits arising from The Network Affect, a phenomenon in which "…the value of a product or service increases according to the number of others using it" (Carl Shapiro and Hal R. Varian (1999) in their book 'Information Rules'). In fact, it's difficult to imagine them enjoying such widespread success if they had been based on a patchwork of proprietary, non-interoperable technologies.

Open source technologies expand on this concept by providing detailed information about the functionality of a hardware or software element, and most or all of the information needed to implement it. Much like how Wikipedia relies on its users to keep its entries accurate and up-to-date, most open source software platforms rely on their user communities to develop applications, bug fixes, and improved versions of the original creation and its associated support tools. For hardware designs, this support also includes mechanical drawings, schematics, firmware, component and material lists, and assembly instructions.

The Network Effect also helps drive the success of open source technologies. Once a platform's user community reaches a certain level, it forms a so-called virtuous circle, whereby the abundance of development resources attracts even more users. In the case of hardware platforms, the growing user base stimulates the emergence of an ecosystem of vendors who compete to offer components, assemblies and accessories at increasingly attractive prices.